

Design of Battery Management System for an Autonomous Underwater Vehicle

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Abstract—Energy Storage systems (ESS) have become an important aspect in the arena of electrical distribution. The capability to monitor, control and optimize the performance of battery modules has become a crucial part of such systems. This paper presents a method to create a Battery Management System compatible with an underwater set-up. The model consists of Lithium-Polymer (LiPo) batteries and uses State of Charge (SoC) estimation method to determine the battery characteristics. It also protects batteries from overcharging, over-discharging and overheating. Further, the BMS design offers passive cell balancing and incorporates swapping in case of over-usage of one of the battery module.

Keywords:- Battery Management System (BMS); Battery Swap; Autonomous Vehicles; State of Charge (SoC); Protection Circuits

I. INTRODUCTION

THE evolving technology of smart Electric Vehicles (EVs), uninterruptible power supplies (UPSs) and other energy systems have contributed towards making battery as the most prominent energy storage device. The power distribution and management is an important task when it comes to underwater systems. The battery technology in such systems is growing at a fast pace, hence it becomes important to improve the battery performance and ensuring that they remain functional throughout their usage. The safe operation of the battery relies on the primary protection features and cell balancing. Subsequently, in the occurrence of unexpected events, systems may require the disconnection of malfunctioning module(s). This battery management reacts to internal or external events, offers high endurance providing a continuous power supply. The BMS consists of two LiPo batteries, consisting of four cells each, powering two different electrical units respectively. The model measures the State of Charge (SoC), current, voltage, temperature and determines the battery state. Whenever required, it issues warnings about faults and takes appropriate actions. The model provides an uninterrupted power supply, by using passive cell balancing. It allows the stack cells to all have the same capacity, which enhances the constancy of the system.

A. The battery modules

It is necessary to decide the type of battery module before designing a BMS. Batteries having similar dimensions, and same chemistry can have different characteristics viz. internal resistance, self discharge rates, capacity, and variation in behaviour due to temperature change. These differences lead to a need for different BMS systems. From the various available battery modules such as the Li-ion, Li-polymer, Lead-acid, Ni-Cd etc., this paper discusses the BMS for a two Li-polymer battery system. This system is used to power an underwater vehicle.

B. BMS design

As mentioned before, the BMS is designed for an underwater vehicle, which has a central board looking after the functioning of all the other components present. It is crucial for the BMS board to communicate with the central board. The design of the BMS comprises a battery management IC (BQ76952), a micro-controller (ATmega328P), and an IC (LTC4421). The BMS architecture for each IC is segregated, and the two are connected together using the LTC4421. The BQ76952 measures the current flowing through battery and calculates SoC (using coulomb counting), voltage of cells, as well as the total voltage of the battery and temperature, and reports all of these values to the micro-controller. It also has several protection systems in place viz. over-voltage, under-voltage, over-current, under-current, over-temperature, under-temperature, pre-charge/discharge, and short-circuit. The IC can detect these issues and stop the supply whenever needed.

Two micro-controllers are used for monitoring the two BQ76952 ICs. The micro-controllers communicate with the ICs to send, collect, and process data, and then execute the tasks accordingly.

Two ICs, also known as High Power Prioritized Powerpath Controllers are used in the BMS each to control the supply to a Power Load. The LTC4421 has an inbuilt swapping management system, which calculates the voltage of each supply and selectively chooses which supply to connect to

the load. The switching of the supply happens in a short span of time i.e. few tens of micro-seconds. It also limits high inrush currents by using sense resistors and MOSFETs.

II. METHODS

A. Battery State

The battery state is the most important aspect in a Battery management system. It mainly corresponds to the State Of Charge (SoC) of the battery.

1) *SoC estimation Methods*: Wen-Yeau Chang [1] and Fathoni et al. [3] discuss and compare about various estimation methods among which some important ones are summarised below:

- **Direct Measurement**: These methods use direct measurement of entities to calculate the SoC. However these methods are only useful when the relationship of OCV with these entities is available.
 - 1) **Open Circuit Voltage Method**: This method uses the complex relationship of open-circuit voltage and SoC in Li-ion batteries (Fig. 1) to calculate SoC. This method is feasible for lead acid batteries owing to their linear SoC-OCV relation.
 - 2) **Terminal Voltage Method**: It uses relationship of terminal voltage with SoC to calculate it.
- **Book-keeping Estimation**: Book-keeping estimation method uses the battery discharging current data as input. This method allows for some internal battery effects as self-discharge, capacity-loss, and discharging efficiency.
 - 1) **Coulomb Counting Method**: This method integrates the current passing through the battery to calculate the value of charge that came out. The Coulomb counting method is performed to estimate the $SoC(t)$ which is estimated from the discharging current, $I(t)$ and previously estimated SoC values, $SoC(t-1)$. SoC is calculated by the following equation:

$$SoC(t) = SoC(t-1) + \frac{I(t)}{Q_n} \delta t$$

- 2) **Modified Coulomb Counting Method**: To improve the Coulomb counting method, a new technique, the modified Coulomb counting method is proposed. The modified Coulomb counting method uses the corrected current to improve the accuracy of estimation. The corrected current is the function of discharging current. There is a quadratic relationship between the corrected current and discharging current of battery, which is given by:

$$I_c(t) = aI(t)^2 + bI(t) + c$$

A better method is discussed by Baccouche et al. in [7]

- **Adaptive Systems**: In the adaptive systems, various neural networks like BPneural network, RBFneural network, and Kalman Filter, and Extended Kalman Filter

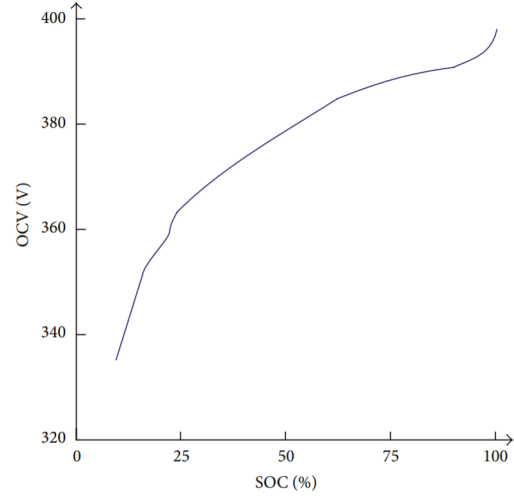


Fig. 1. The State of Charge (SoC) vs Open Circuit Voltage (OCV) curve for a Li-ion battery to explain the linear relationship. The curve can be approximated as a straight line for an approximate voltage range

etc. are employed to obtain a more accurate estimate of SoC. These systems are discussed in detail by J. Meng et al. in [8].

2) *SoH (State of Health)*: It is an important parameter to understand the state of the battery. It is defined as the maximum charge available in the battery as a percentage of the maximum charge that can be stored in the battery:

$$SoH(\%) = \frac{C_{max}}{C_{rated}} \times 100$$

where C_{max} is maximum charge available in the battery and C_{rated} is maximum charge that can be stored in the battery.

3) *Types of battery management ICs*: Based on complexity, there are three types of battery management ICs available [9] -

- 1) **Battery Protectors**: This is the least complex hardware for battery management. It has simple protection features like over-voltage, and under-voltage protection, over-current and under-current protection, over-temperature and under-temperature protection short-circuit protection.
- 2) **Battery Monitors**: These have all the features necessary for the safe working and longevity of the battery. It measures individual cell voltages for cell balancing to increase battery runtime and battery life. All the protections have flexible thresholds. A robust communication system sends and receives data to an MCU for a more personalised use.
- 3) **Battery Gauge**: It has all the features of a battery monitor along with encryption and authentication of battery and dynamic calculation of SoH for a more healthy battery life.

We used a battery monitor because it met all the requirements for a vehicle. We have also implemented several additional features such as the option to manually calculate SoH in the MCU itself for compactness and robustness in the system's functioning.

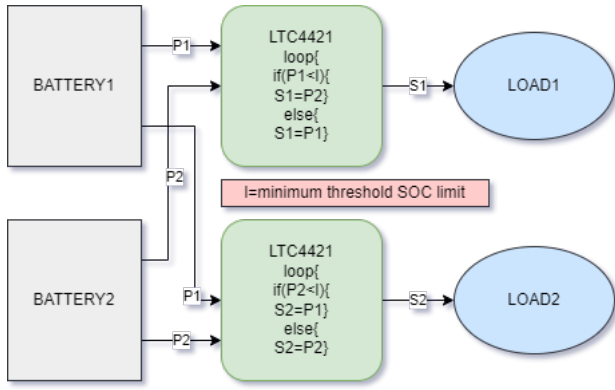


Fig. 2. The mechanism for swapping of batteries. Here, two individual batteries (which are responsible for independent functions) are connected through LTC4421 such that any battery reaching a voltage level less than threshold results in shifting of the loads

4) *Battery Swapping*: We have two batteries with 4 cells each in the BMS. One battery is primarily used to power the thrusters, while the other one is used for the electrical components. Since both batteries have different loads, there is a substantial probability of a significant difference between the SoC of the two batteries. As can be seen from the cell balancing example, it is clear that if *battery*₁ gets discharged first then the charge in *battery*₂ cannot be used and eventually the vehicle needs to be stopped. To solve this problem, a battery-swap system has been implemented in the BMS.

We have used the LTC4421 which senses the voltage of two batteries and compares them with the threshold levels. If the voltage of *Battery*_{*i*} goes below the specified threshold limit, then both the loads are connected to the *Battery*_{*j*} s.t. *i* ≠ *j* without disconnecting the *load*_{*i*} for a significant amount of time. (Fig. 2)

5) *SoC-OCV relation*: For a BMS, it is necessary to ascertain the value of OCV of the battery at any instant. To find this, we need the OCV vs SoC relation which may be used to find OCV from a known SoC value.

Since OCV of the battery cannot be measured directly, we use pulse charging¹ i.e. as discussed in [2] by Chin et al. During the test, the battery was first charged with 15 A to the fully charged state in constant current-constant voltage (CCCV) mode to around 3.5 V. Then by discharging the battery cell at a pulse current 15A for 15 s, relaxing for 300s, discharging for 1200 s and then relaxing again for 1 hour until the terminal voltage reached the cut-off voltage of 2.5 V. By using MATLAB's best curve fitting tool it was found that the following relation held:

$$V_{oc} = 3.1568 - \frac{0.011}{SoC} - 0.0035SoC + 0.0025\ln(SoC) - 0.0397\ln(1 - SoC)$$

¹Pulse charge/discharge is a periodic current square wave. Researches and practices have proven that both the charging efficiency as well as cycle life can be significantly improved using pulse function.

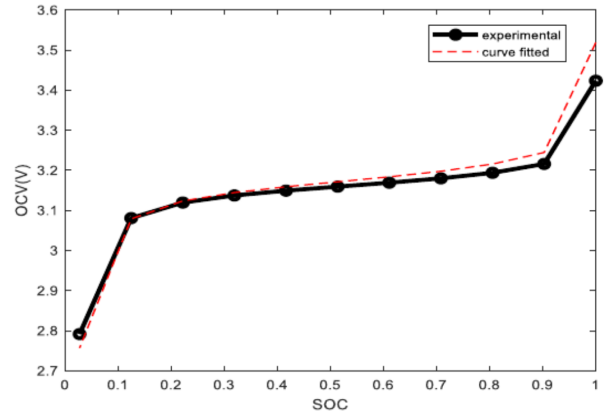


Fig. 3. The Open Circuit Voltage vs State of Charge relationship. Here, the black colored curve displays results obtained through an experimental dataset and red colored dotted-line curve is obtained after employing curve-fitting for a Lithium Polymer battery

B. Battery Health

For maintaining a healthy battery performance, some points need to be considered:

1) *Cell balancing*: Cell balancing [4] plays a significant role in increasing battery runtime and battery life. There are various algorithms discussed, among which two are the most important:

- **Active cell balancing**: When the SoC of cells of a battery becomes unequal, the charge needs to be redistributed among the cells, i.e. transfer charge from a higher SoC cell to a lower SoC cell. This process is called active cell balancing [5][10]. More explanation can be found in Fig. 4
- **Passive cell balancing**: When the SoC of cells of a battery becomes unequal, we discharge the higher SoC cell on an external load. This process is termed passive cell balancing [6]. More explanation can be found in Fig. 5

Consider an example of a battery *B* having *n* cells *C_i*

Experiment 1: While discharging, cell *C_i* has SoC = 50% and *C_j* has SoC = 30% and minimum threshold SoC is 30%. Then without cell balancing, the vehicle will stop working and charge in the *i*-th cell will be wasted. Instead, if active cell balancing is used, then both the cells will have a SoC = 40% which would result in increased runtime.

Experiment 2: While charging, cell *C_i* has SoC = 100% and *C_j* has SoC = 90%. Then without cell balancing, charging would stop, and the capacity of the *j*-th cell will not be fully utilised. But if passive cell balancing is used, then the *i*-th cell will start discharging dynamically in an external load, till both cells get fully charged.

We use passive cell balancing as the BQ76952 has inbuilt system for this. The IC measures voltage of each cell and uses differential voltages to monitor and implement cell balancing.

2) *Protections*:

- **Voltage/Current/Temperature Protections**: If the battery

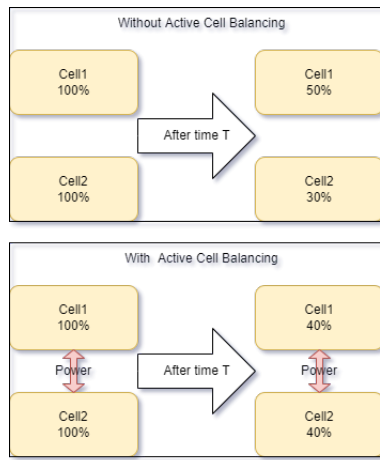


Fig. 4. Battery cell performance with and without active cell balancing. Active cell balancing charges all the cells equally by discharging higher SoC cells and charging lower SoC cells. Without cell balancing, the cells continue to have a difference in their SoC

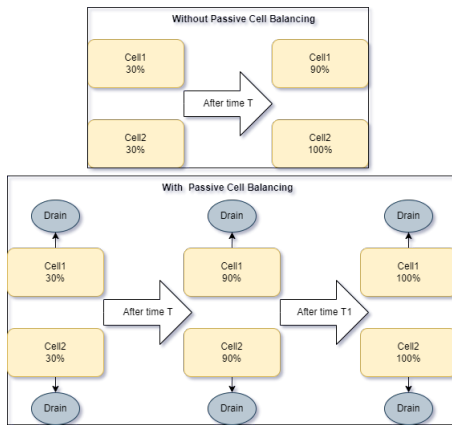


Fig. 5. Battery cell performance with and without passive cell balancing. Passive cell balancing allows the cells to have equal SoC by discharging the cells with higher SoC to a level equivalent to lowest SoC cell. Without cell balancing, the cells continue to have a difference in their SoC

gets charged more than its maximum threshold voltage, or during the discharge, if its voltage falls below the minimum threshold limit, then its inner components may get damaged. Similarly, passing a very high or a very low current, or operating it under adverse temperature conditions may decrease the battery life or even damage the battery. To overcome this difficulty, the BQ76952 IC integrates the inbuilt protection systems, which allows for manually setting of the threshold limits with the help of MCU or BQ-studio. For temperature protection, external thermistors can be added to measure the temperature of the components far from the battery. Along with this, the IC has an inbuilt short-circuit protector.

- Pre-charge/discharge Protections: When the battery is switched off, and is plugged to charge or connected to a load, it is advisable to increase the flow of current gradually, and not instantly. So, the BQ76952 IC has

a Pre-charge/discharge protection, which regulates the amount of current passing through the battery.

C. Communication

The BQ76952 IC supports communication over UART (Universal Asynchronous Receiver-Transmitter), I^2C (Inter-Integrated Circuit), SPI (Serial Peripheral Interface), and HDQ(High-Speed Data Queue). However, I^2C is the primary mode of communication with the MCU. UART is used as a backup system to directly communicate with the On-board Computer (Serial Board computer, SBC) in case of failure. The MCU communicates with the SBC through CAN.

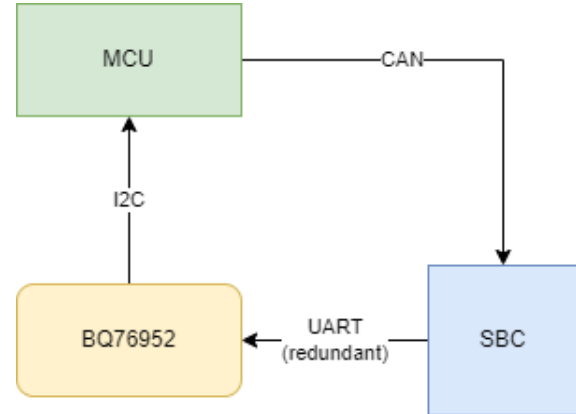


Fig. 6. Communication of BQ76952 with central board of the under-water system. As the Micro-Controller Unit (MCU) of the central board communicates with the on-board computer or Serial board computer (SBC) over Controller Area Network (CAN) protocol, the battery management IC BQ76952 communicates with MCU via I^2C . There is also a redundant communication between on-board computer and BQ76952 over UART

III. RESULTS AND DISCUSSION

A. Battery Swapping

Since battery swapping is a crucial factor in implementing a BMS, simulations need to be performed to check the time the LTC4421 (Swapping IC) takes to change the supply for the load.

The simulation parameters are given below:

- $Supply_1 = 16V$
- $Supply_2 = 12\exp(-t/25)$ where t is in seconds.
- Minimum threshold limit of output= 6.05V

The simulation outputs are given in Fig. 7 and Fig. 8

Time calculated to switch from $Battery_1$ to $Battery_2$ is 45μ seconds.

B. Temperature measurement

Temperature measurement is a crucial task to be done in a battery management system because if the high temperature can affect battery health adversely, it also decreases the efficiency of battery and may damage some electrical components of the circuit.

For a very accurate measurement of temperature, polynomial temperature mapping can be used. In this method, resistance

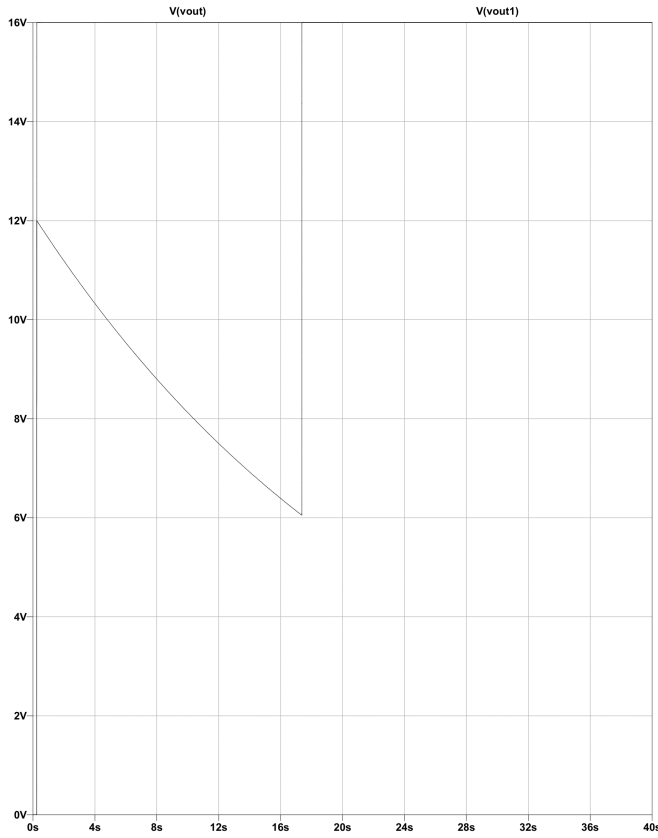


Fig. 7. Voltage vs time for both loads of a simulation in LTSpice Simulation a battery swapping system in LTSpice having two batteries with supplies constant 16V and 12V exponentially decreasing with time constant 25 seconds and minimum threshold voltage 6.05V. As shown in Fig. 2 the connections are made and output is observed at both loads

measured is given as a quadratic polynomial of temperature. i.e.

$$R = a + bT + cT^2 + dT^3 + eT^4$$

The constants a, b, c, d, e can be calculated by using linear regression with T, T^2, T^3, T^4 as variables.

We calculated the constants for the thermistor 103AT-2 Sematic and found the following results: $a = 2.447e + 04, b = -1321.1498, c = 39.7933, d = -0.5269, e = 0.0024$ The resistance vs temperature graph obtained is given in Fig. 9:

IV. CONCLUSION

A battery management model for an underwater system is presented in this paper which can find its application in Autonomous Underwater Vehicles, hybrid electrical vehicles, Remotely Operated Vehicles, etc. The battery shift method proposed to increase the endurance of the vehicle can be extended to a number of batteries. The future novel still looks for a method a to swap batteries without affecting the functioning of the electrical system.

The further research will revolve about analyzing more advanced methods to measure SoC using neural networks and other estimation methods [11]. The network is trained with voltage and current values as input and SoC as output. It is then tested for various external parameters like temperature,

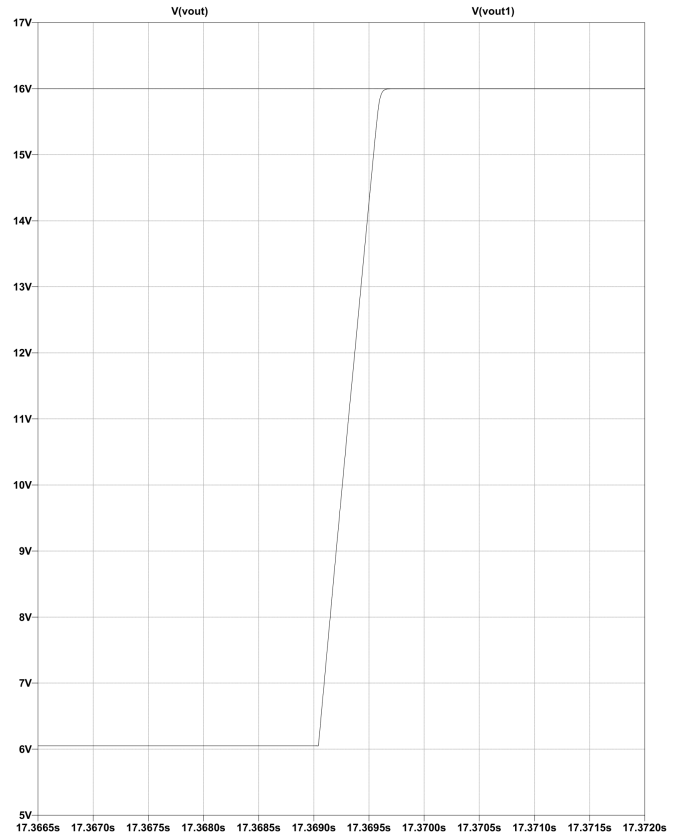


Fig. 8. Zoomed in graph of Fig. 7 at the time of swapping

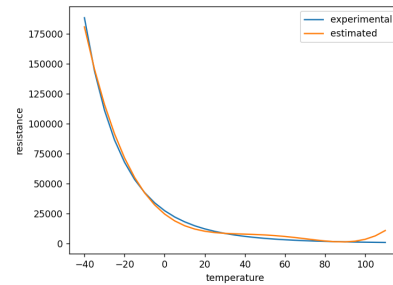


Fig. 9. Polynomial temperature model: A regression graph of bi-quadratic temperature-resistance relationship of Semitec AT103-2 thermistor. The blue curve represents experimental data while orange curve is regressed data.

run-time, load etc. This method will improve the accuracy of SoC.

Currently, if the first battery drains faster than the second then both the loads will be connected to the second battery, when first battery reaches its minimum threshold. In future, the system can be constructed to be more dynamic i.e. if the first battery drains faster than the second then both the loads will be interchanged and hence improve the battery life or hot-swapping can also be used i.e. if n batteries are needed to run the vehicle then $n + k$ batteries should be used. If any battery malfunctions or gets discharged faster the load can be transferred to fully charged working batteries. It will ensure that the vehicle is robust and will improve battery life.

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